

1 Discussion

The paper asked whether the method developed for PA induction (a cautious verifier over a class of templates, sound at all times and exact once the data contain an anchor) also learns a universal axiom from its instances, each $\text{ZF}(C)$ schema from its instances, and many axioms and schemas at once without labels. **??????** give the answers, and **??** tests them. This section says what they mean for the larger project they belong to, learning which inference rules a community uses from imitation, coherence and the world (Claude (Anthropic), 2026) (the *parent report*). It then relates them to earlier work, collects the open problems, and states limitations and scope. It claims no new results.

1.1 What the answers mean

An instance schema is not a universal sentence. Data $\varphi(t)$ at closed terms determine the instance schema $\varphi(z)$; two instances whose terms have different head symbols already do (**????**). They do not determine the sentence $\forall x\varphi$. The passage from all closed instances to $\forall x\varphi$ is an ω -rule step. It is truth-safe in the intended structure for every φ iff the closed-term substructure is elementary (**??**): in $\mathbb{N}$, but not in $\mathbb{R}$, in V or in a nonstandard model of PA. It is not derivable: Q proves each $0 + \underline{n} = \underline{n}$ but not $\forall x (0 + x = x)$ (**??**). So closed data support two targets, the closed-instance schema $\text{inst}_c(\varphi)$ and the open-instance schema $\text{inst}_o(\varphi)$, which in closure-normal form is the axiom. A cautious learner whose class admits parameters as values takes the larger one after the first anchor, silently (**??**, **??**). Which one a community uses is a modelling decision that closed data cannot settle. A closedness guard encodes one answer, a single datum at a parameter the other (**??**). Where $\forall$ -elimination is routinely applied to variables, as in proof assistants, the second reading is the realistic one and the gap disappears. Without labels, the universal sentence is reached differently, as a merge of instances that the world does not refute (**????**). What certifies it then is the absence of a counterexample, not a derivation, so its truth-soundness is only as good as the oracle; $\text{R-}\Delta_0$ suffices for quantifier-free φ in arithmetic (**??**).

First-order learnability depends on the formulation. Whether plain first-order anti-unification learns a schema is a property of how its instances are written, not of the axiom. A pattern schema with one metavariable is a (guarded) first-order pattern iff the metavariable is applied to literally the same argument tuple at every occurrence in the chosen syntax: the same names, the same de Bruijn indices, or, when names vary, the same binders (**????**). So $\in$ -induction is first-order in de Bruijn notation but not with names, Collection and $\exists!$ -Replacement the other way round, and α -renaming the frame changes the verdict (**??**). Where the criterion fails, no finite union of guarded first-order schemas covers the schema without a false member (**??**). The one exception, spelled-out Replacement with textbook names, is a truth-sound over-generalization that no refutation can expose (**??**). PA induction shows the same dependence on a larger scale. Raw instances defeat every finite union of first-order schemas (**??**), while the Tarski-style form IndEq and the Sub encoding make induction a single first-order pattern with the same anchor events (**????**). For a checker the lesson is to learn in a class whose verdicts do not depend on choices that a community regards as notation. The λ convention of **??** does this: bound-variable names do not matter, freshness side conditions are built in, and capture cannot occur. In a first-order syntax capture can make an unguarded learner unsound outright (**??**).

Why the pattern fragment matters. Every ZF schema, in every formulation considered, is a higher-order pattern (Miller, 1991): its metavariable is applied only to distinct bound variables (??). The reason is structural. $L_{\in}$ has no function symbols, so an instance of a ZF schema only renames variables inside the body. In this fragment existing technology suffices. The pattern lgg exists, is unique, and recovers each schema from any anchor (??). For one metavariable the anchors are root variation plus non-vacuity in every class between PAT and SO° , so the choice of class does not matter (??). It matters again with several metavariables, which is how an untagged mixture behaves: then a Plotkin-(D)-type event appears, it depends on the class, and DT° needs more diversity in the data than PAT (??). An anchor is always an anchor *in a class*, and a method should state its anchor theorem for the class it uses.

Induction needs derived occurrences. Induction is our main schema outside the pattern fragment. In T_{Ind} the occurrences $P(0)$ and $P(Sx)$ apply the motive to terms, so their values are *derived* from the value at the pattern occurrence $P(x)$. First-order anti-unification is unsound on two instances (??), some of its unsound lggs are never refuted by closed truths (??), and pattern anti-unification is unsound as well (??). The determinate class DT° is built for this case. Requiring a pattern occurrence for every metavariable is what keeps matching unique and linear (??) and the set of minimal covering templates finite (??). Dropping the requirement (SO°) needs stronger anchors (??); allowing metavariables inside arguments makes the set of minimal covering templates infinite (??). The price of derived occurrences is non-unitarity: before an anchor the version space can have several minimal elements, up to 4^n (????). Caution absorbs this price. The verifier intersects all minimal templates, which is a polynomial feature test (??), and on the pair behind K_0 the intersection rejects the false sentence that the K_0 -shaped minimal template contains (??). On anchor data the least covering template exists and is the target (??). Determinacy also matches practice: proof assistants record or recompute the motive, and in DT° the motive is the unique matcher, so for closed $\forall$ -form steps recording it changes neither anchors nor rates (??).

Refutation is the source of labels. Without labels the learner must decide which instances belong together, and positive data cannot decide it. Without a bound on the number of schemas the cautious learner accepts only its data, even with sound negatives (??). With a bound, spare slots are paid for in data diversity that no sound negative evidence replaces (??, ??). The two kinds of evidence guard against opposite errors. Refuted sentences remove over-general *lumps*. A bound removes *splits*, which are sound and therefore invisible to refutation. DTRC needs no bound: it merges two clusters whenever their union has an unrefuted minimal template, and stops only when no merge is possible. Under refutation separation this recovers the hidden labels, and within each cluster the learner *is* the tagged learner, with its anchors and rates (??). With the exact bound and anchor-pair separation, the learner generates the negatives it needs itself (??). In the parent report the world and coherence channels exclude unsound rules and locate faulty steps. Here they also play the role of citations, which formal proofs record and informal proofs often leave implicit. For PA, ZF and ZFC they separate the axioms at small depth, with Russell’s paradox doing the work exactly where the axioms look alike (??). Two limits are intrinsic. No computable learner can eliminate the refutation depth (??). And compression does not replace refutation: MDL tracks how often schemas are used, not where their logical boundaries lie (??).

Truth-soundness and target-soundness come apart. “Which axioms does this community use?” asks for target-soundness; “is it safe to accept this step?” asks for truth-soundness. Refu-

tation serves only the second. A merge whose templates have no false instance is never refuted (??). Learning $\forall x\varphi$ from its instances, absorbing a redundant axiom into a schema, and the valid residual templates of the experiments (??) are merges of this kind. They make DTRC truth-sound but not target-sound (??), and the loss is unavoidable: on some clean data with anchors no learner is both exact for one valid practice and target-sound for another (??). Target-soundness without separation needs evidence other than the world: labels, a known bound $k \geq k'$ (intersecting with the k -union learner restores it, ??), or the community’s own judgments. Truth-soundness, in turn, holds only relative to a residue of unrefuted unsound merges (??), and the residue has natural members: the universal-set schema obtained from weak forms of Union and Power, which only coherence refutes (??); the K_0 templates of induction (??); and, with mistakes in the data and a budgeted refuter, a false sentence that the implementation actually accepted (??). This is a sharp case of the general picture of the parent report: every signal is one-sided, and what can be proved is soundness together with a declared residue.

1.2 Relation to the literature

Anti-unification and inductive logic programming. The first-order core is the least general generalization of Plotkin (1970) and Reynolds (1970), the basis of bottom-up generalization in inductive logic programming (Muggleton, 1991). Plotkin’s witness events (R) and (D) are our anchor conditions for first-order targets, and the larger class DT° has the same anchors on such targets (??). The feature test of ?? is the second-order form of Plotkin’s rule that equal columns get equal variables. In meta-interpretive learning the shape of clauses is fixed by metarules, second-order Horn clauses given in advance (Muggleton et al., 2015; Cropper and Tourret, 2020); in our setting the second-order templates are what is learned.

Higher-order anti-unification and matching. For higher-order patterns the least general generalization is unique (Pfenning, 1991) and computable in linear time (Baumgartner et al., 2017); the skeleton template $T_0(D)$ of ??, with columns merged up to renaming, is that lgg. Second-order generalization beyond patterns is studied by Hirata et al. (2004) and in the frameworks surveyed by Cerna and Kutsia (2023); we have not checked whether DT° or the feature test appears there. DT° extends patterns by derived occurrences: it gives up unitarity, but the intersection of the minimal generalizations stays polynomial (??). On the matching side, second-order matching has finitely many matchers (Huet and Lang, 1978), and its NP-hardness (usually attributed to Baxter’s thesis, which we have not checked) needs arguments that contain metavariables. DT° matching is unique and linear, the case of Miller’s pattern unification with one side ground, followed by evaluation (??). The closest relatives we know of are the *deterministic second-order patterns* of Yokoyama et al. (2004), which relax Miller’s restriction while keeping matching deterministic and fast (Niederhauser and Middeldorp, 2026 give a unification procedure for them), and the functions-as-constructors fragment of Libal and Miller (2016). We know the first class only from its abstract and secondary descriptions, according to which every argument of a metavariable must contain a bound variable, which $P(0)$ does not. We have not compared either class with DT° , nor checked whether anti-unification has been studied for them.

Learning from positive data. Gold’s theorem (Gold, 1967) is why an untagged learner needs a bound, or a restriction on the practice such as refutation separation (??). Every anchor is a tell-tale in the sense of Angluin (1980b), and up to effectivity anchors are the $\subseteq$ -tell-tales of strong-monotonic learning (Lange and Zeugmann, 1992). First-order patterns are the tree

analogue of Angluin’s pattern languages (Angluin, 1980a). Finite elasticity and its preservation under bounded unions (Wright, 1989; Motoki et al., 1991) give anchors for unions of determinate templates (??); Shinohara (1991) shows that languages defined by length-bounded elementary formal systems are learnable from positive data, and Arimura et al. (1994) find minimal multiple generalizations for unions of pattern languages (we have not checked how their complexity results relate to ??). What this paper and its parent report add to that tradition is soundness at all times against adaptive provers rather than convergence alone, explicit anchors by witness events with rates, and refutation by the world as negative evidence, which under separation lets the learner find the number of schemas.

MDL and the size principle. Minimum description length (Rissanen, 1978) and the size principle (Tenenbaum and Griffiths, 2001) prefer hypotheses under which the data are probable. For schemas this tracks the statistics of usage: a naive two-part code splits induction by the main connective of the motive, by a margin linear in the number of data, and under a natural usage law even depth-aware adaptive codes split once the data are numerous (??). The cautious verifier answers a different question. It accepts what every consistent hypothesis accepts, and it accepts a whole schema only after an anchor. The refutation test controls the over-general merges that compression can favour, and the anchor condition controls the splits.

Axiomatizability by schemas. Vaught (1967) studied axiomatizability by a schema: a formula with a new relation symbol whose substitution instances axiomatize a theory. He showed that every recursively enumerable theory that directly interprets a weak set theory is axiomatizable in this way. In present terms such a schema is a second-order template with one formula metavariable, induction being the paradigm, and DT° is a version of the notion with unique matching, for which the schema can be learned from its instances. The theorem of Ryll-Nardzewski (1952) that PA is not finitely axiomatizable concerns consequence and uses nonstandard models. Our impossibility for first-order schemas (??) concerns the syntax of substitution and uses only the refutability of a few sentences in $\mathbf{Q}$. Neither implies the other: PA is $\mathbf{Q}$ plus one guarded first-order pattern (??).

Proof assistants. Formal libraries record what the tagged setting assumes, namely which axiom, lemma or recursor a step applies. They also record or recompute the motive of an induction. In Lean 4 the recursor takes the motive as an implicit argument, and the `induction` tactic produces a kernel term that records it as a λ (de Moura and Ullrich, 2021); in our terms this is the DT° matcher. Metamath’s `set.mm` states induction with implicit-substitution hypotheses and distinct-variable conditions (Megill and Wheeler, 2019); in our terms this is a first-order pattern with freshness guards, and `set.mm`’s conditions, read as “not free”, are sound but not minimal (??). So for formal mathematics the tagged results of ?????? are the relevant ones. The untagged theory of ?? matters where citations are missing, as in informal mathematics.

1.3 Open problems

The research records list the following open problems; gray tags name the record, and pointers give the result that bounds each problem. Two problems listed in the records have since been settled: finiteness of minimal covering templates in DT° (??) and the complexity of the refutation test (NP-complete, ??, with the restriction in O20 below still open).

One template.

- O1 The exact anchor condition without (Rich). With (Rich) it is $(R^*)+(N)+(U)$ (??); without it the condition is strictly weaker single §13.
- O2 Syntactic versus semantic generality: does $\text{inst}(T) \supseteq \text{inst}(T')$ imply $T \succeq T'$ in DT° ? The verifier does not need it; the analogue for first-order schemas with term metavariables holds under a richness assumption (??) single §13.
- O3 Equation kills: is their number linear (??)? Escalations are $\Theta(N^2)$ in general (????) and at most $2N$ without binders (??). The exact escalation counts for the ZF schemas are not known either single §13; cases §6.
- O4 Enumeration of $\text{Min}(D)$ with polynomial delay, and counting it; $|\text{Min}(D)|$ can be 4^n (??) single §13.
- O5 Size-bounded subclasses of DT° . Induction is closed in DT_s° iff $s \geq 12$, and for $7 \leq s \leq 11$ the verifier accepts a false sentence on every anchor (??); a general theory is missing single §13; prior induction C6.2.
- O6 Lower bounds for the coincidence terms of the rates when the data law has infinite support (??) single §13.

Other classes and formulations.

- O7 Anchors in SO° with several metavariables: is root-level distinctness necessary? Conjectured not (??) cases §6.
- O8 The exact SO° anchor condition for universal axioms over languages with closed terms, an analogue of the shielding condition for induction (??); only counterexamples are known (??) cases §6.
- O9 How DT° relates to deterministic second-order patterns and to the functions-as-constructors fragment (Section 1.2) prior induction §3.
- O10 Does ZF without Foundation prove Collection? By ?? this decides whether the named lgg of spelled-out Replacement is truth-sound over that theory. We do not know the answer; it may be known cases §6.
- O11 Communities that α -rename only some binders of a schema, or rename inconsistently across a corpus; ?? covers full renaming only cases §5, §6.

Many schemas without labels.

- O12 Spare-slot thresholds beyond one-variable schemas: binary function symbols with more spare slots than head classes, several metavariables, templates with parameters, and T_{Ind} with eight or more slots (??) untagged §10.
- O13 Usable bounds on the separation depth for natural targets. No uniform computable bound exists (??); for PA, ZF and ZFC the depth is small (??) untagged §10.
- O14 A natural permanent residue, i.e. a merge of natural axioms that is unsound and never refuted. In arithmetic it needs quantifiers in the template (??) untagged §10.
- O15 A merge rule with a theorem that neither mistakes nor sound merges fragment a target's anchor; by ?? it needs more than anchors in the data (??, ??) untagged §10.

- O16 A theorem that some natural universal code selects the target partition under a well-specified usage model, and a characterization of when usage heterogeneity makes it split (??) ^{untagged §10}.
- O17 Guarantees for bootstrapped coherence when only some clusters are pure (??) ^{untagged §10}.
- O18 Refutation separation for similar targets: characterize, in terms of the targets alone, when every cross merge is refutable. Genuine schema pairs can have both refutable and valid merges (??, ??) ^{experiments §6}.
- O19 A guarantee for the sharing pass without its purity hypothesis (??) ^{experiments §6}.

Cost and oracles.

- O20 Is the refutation test of DTRC still NP-hard when the data are true and the negatives are false in a fixed structure, as in DTRC? The hardness proof of ??(d) uses a false datum ^{single §13}.
- O21 The exact union verifier under realistic restrictions, e.g. bounded binder depth with fixed $k \geq 3$ (??) ^{single §13}.
- O22 A natural refuter that refutes every false minimal template of a cross pair within a budget polynomial in the data. With mistakes in the data, budgets 80 and 400 did not suffice (??) ^{experiments §6}.
- O23 World oracles that construct witnesses from generic objects or terms, which would refute comprehension failures such as $\forall a \exists b \forall x (x \in b \leftrightarrow a \in x)$ (??) ^{experiments §6}.

Beyond the records. The limitations below suggest further directions, which the records do not analyse: templates over typed object languages (the motive of Lean’s recursor is dependently typed); inference rules with premises and with contexts, which enter here only through encodings such as Sub and Metamath’s; and real corpora, such as set.mm or a Lean library, as data.

1.4 Limitations

Interpretation. We read the first question’s “sentences of the form $\varphi(x)$ ” as instances at terms. Read literally, with x free, such a datum is in closure-normal form already the axiom (??). *Languages and data.* Object languages are single-sorted first-order languages; the examples are arithmetic and set theory. Data are in closure-normal form, and the theory treats parameters as constants, while the implementation aligns them (??). The practice is assumed correct except in the noise analysis (??), and a frequent systematic error cannot be told from a rule. Rates assume i.i.d. data; real usage is not i.i.d., and ?? shows that its heterogeneity matters. *Classes.* DT° excludes metavariables inside arguments, and the implemented DT_F° also excludes term metavariables of positive arity; the feature test of ?? is not implemented. *Oracles.* The refutation oracles are sound but incomplete. The implemented refuter is budgeted and history-dependent, so probe counts are lower bounds on unsoundness (??). Refutation separation is proved for the experimental mixtures only, and the evidence for similar targets is narrow (??). *Experiments.* The data are synthetic and small, with random bodies; no real corpus was used. *Verification.* Each research record was checked by an adversarial referee with independent code, and the records’ corrections are kept (??). Results added in revision, after a record’s referee round, were rechecked by computation and for this paper, but not by an independent referee; among them are the complexity results of ??, ???? and ??. Bounded searches are evidence, not proof. Textbook wordings of the ZF axioms are recalled, not checked against the books, and possible overlaps with the work cited in Section 1.2 as unchecked remain unchecked.

1.5 Scope: checking, not proving

Everything here concerns checking. The learner decides whether a proposed sentence is licensed by the axioms and schemas that a community has been seen to use. It never searches for proofs; the prover of the protocol is arbitrary and may be adversarial (??). The guarantees are of two kinds: soundness against every prover at all times, relative to the class and, without labels, to a residue; and exactness after an anchor, with rates. The costs are data costs and verification costs (linear matching, a polynomial feature test; in the worst case coNP-complete exact union verification for $k \geq 3$ and an NP-complete refutation test), not search costs. Nothing here makes a prover stronger. An accepted sentence is one that the community’s practice licenses, which can differ from what is true (Section 1.1). Possible uses, not tested here, are auditing which axioms a body of informal or uncited proofs relies on (for instance Collection rather than Replacement, or Choice), and flagging for human review the steps that no learned schema licenses, which are the escalations of the protocol.

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